Exploring Skill Requirements in Data Science Job Postings Using NLP-Based Clustering

1. Introduction

With the rapid development of data science and artificial intelligence, demand for data-related roles has increased significantly across industries [1]. However, as technologies evolve quickly and organizations adopt data practices in different ways, job titles such as *Data Scientist*, *Data Analyst*, and *Data Engineer* often correspond to different skill requirements across companies [2]. This inconsistency creates confusion for job seekers attempting to understand which skills are required for specific career paths.

Prior research has explored the use of machine learning and natural language processing (NLP) techniques to analyze job postings. For example, Işığıçok et al. (2023) analyzed job advertisements using TF-IDF and word clouds to identify common skills required for data scientists [3]. While this approach provides useful descriptive insights, it does not differentiate between distinct sub-roles within data science, such as data engineering and data scientist. More recently, Lukauskas et al. (2023) demonstrated that modern sentence embeddings combined with HDBSCAN can be used to generate interpretable job profiles and to provide deeper insights into skill demand within the labor market. [4] However, this paper analyzed the job market in Lithuania in general and did not target any specific field.

Motivated by these limitations, this project applies unsupervised NLP-based clustering and exploratory analysis on job skill requirements to identify meaningful job categories within data-related roles and to differentiate the skill sets associated with each category. The findings aim to support job seekers by clarifying which skills are most relevant for different segments of data-related roles.

2. Analytical questions and data

2.1 Analytical Questions

This project aims to summarize and extract actionable insights from job posting data in order to support students and job seekers in the data science field. To achieve this objective, the analysis addresses the following research questions:

Q1: What core skill sets are commonly required across all data-related roles?

Q1: Can job postings in data science be meaningfully clustered into distinct job categories using SentenceTransformer embeddings and HDBSCAN clustering?

Q2: What skills characterize each identified job category within data science?

2.2 Data

The analysis uses the Kaggle dataset *"Data Science Job Postings & Skills (2024),"* which contains 12,217 job postings collected from LinkedIn [5]. The dataset includes job location, company name, job title, description, and a structured list of required skills. The relatively large sample size enables generalizable observations about skill demand in the data science job market. In particular, the presence of a pre-extracted job_skills column facilitates more accurate and efficient skill-based analysis.

Nevertheless, the dataset has several limitations in relation to the project objectives. First, as the data was collected in 2024, it may not fully reflect the most recent shifts in labour market demand. Moreover, approximately 84% of job postings originate from the United States, meaning the dataset primarily reflects the US job market rather than the global data science landscape.

3. Analysis

3.1 Data Preparation and Exploration

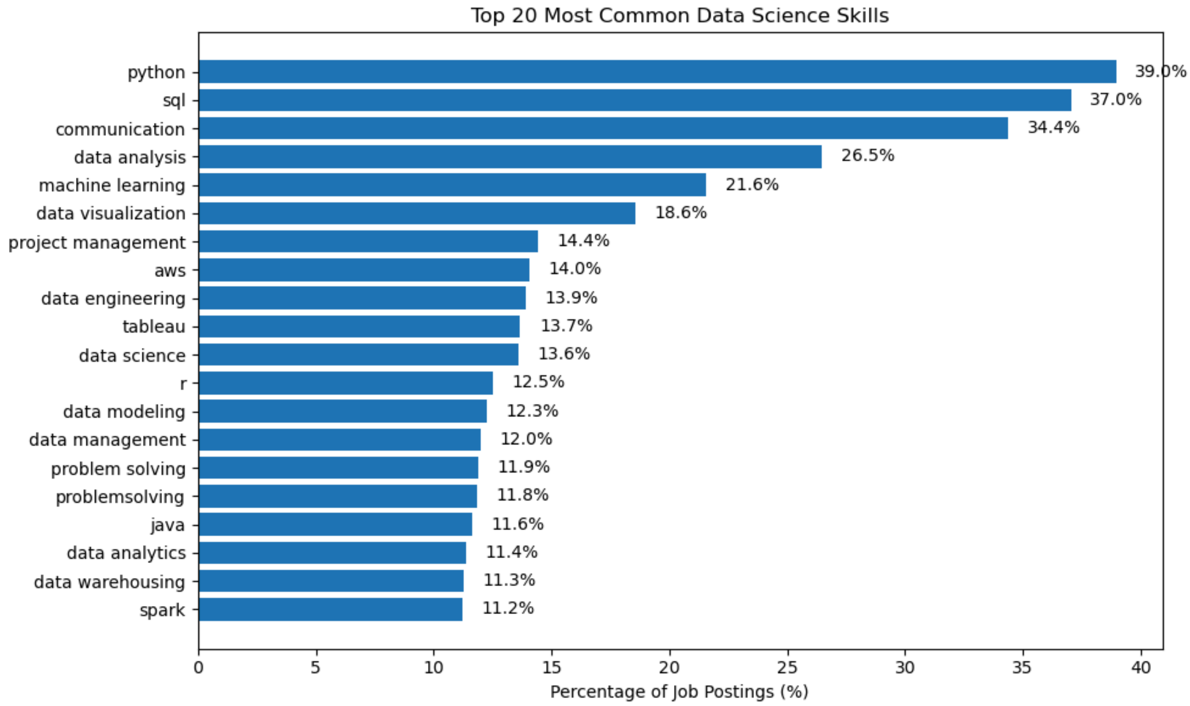
Data Cleaning

Initial data cleaning steps were applied prior to analysis. Job postings without specified skills were removed. The job_skills column contains both technical and soft skills separated by commas, eliminating the need to extract skill-related keywords directly from job descriptions. However, further preprocessing was required to standardize keywords and reduce noise.

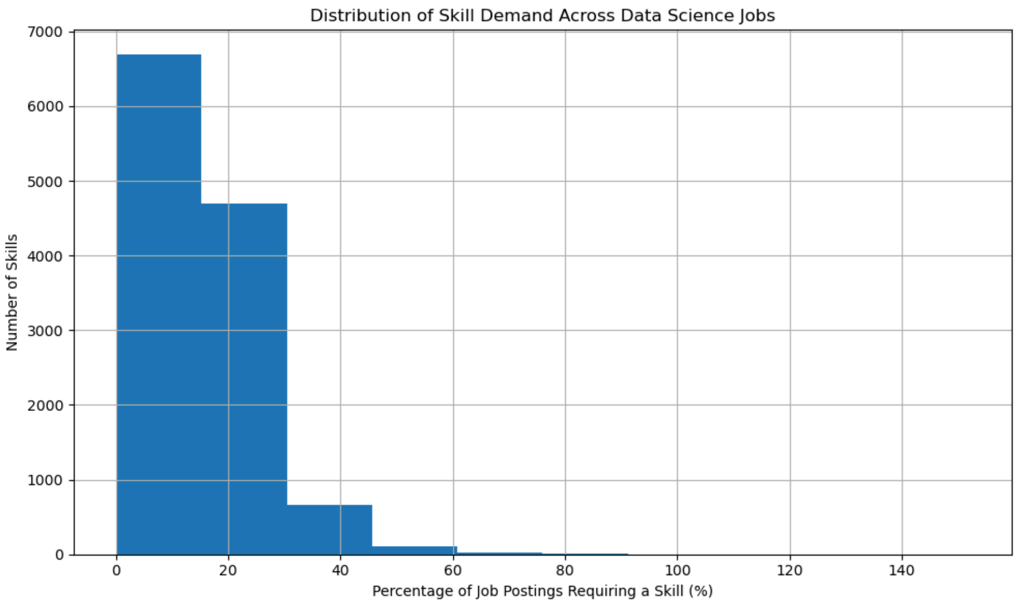
Since the data consist largely of domain-specific terminology, conventional natural language cleaning techniques were not fully appropriate. Instead, manual inspection and domain knowledge were used to guide preprocessing decisions. All keywords were converted to lowercase, and special characters were removed. Generic filler terms (e.g., *ability_to, linkedin, shifts, rooms*) and soft-skill-related keywords that did not contribute meaningfully to role differentiation (e.g., *adaptability*, *priorities*) were removed. Additionally, unrelated acronyms were filtered out.

Data Exploration

In this section, we explore the general patterns of the dataset by analysing the frequency of skill keywords across job postings. The below chart presents the top 20 most common skills across data science job postings. Programming and analytical skills dominate the ranking, with *Python* (39.0%) and *SQL* (37.0%) appearing in the largest proportion of postings, followed by *communication* (34.4%) and *data analysis* (26.5%). This indicates that data science roles require a combination of strong technical foundations and communication skills.



On average, a single job posting requires around 16 different skills. However, the distribution is highly skewed, with some roles requiring a larger number of skills, up to 60 or more in extreme cases. This variation suggests that while many data science roles share a common core skill set, certain positions demand a broader and more specialized combination of skills, further motivating the use of clustering methods to uncover meaningful job categories.



3.2 Feature engineering

Following initial preprocessing, remaining keywords were lemmatized using spaCy. This process reduced the number of unique keywords by more than 10%, improving feature consistency.

To transform the cleaned skill lists into numerical features, two vectorization approaches were considered:

- TF–IDF explicitly emphasizes skills that are distinctive to subsets of job postings by down-weighting common skills such as *Python* and *SQL*. This property is particularly desirable when the objective is to differentiate sub-roles in data science.
- SentenceTransformer embeddings were also evaluated due to their strong performance in prior job market segmentation studies [4]. This approach converts each skill list into a dense semantic representation with 384 dimensions and has the advantage of capturing relationships between semantically related skills.

Both representations were therefore tested empirically in combination with HDBSCAN clustering to assess which approach produced more stable and interpretable clusters. While TF–IDF offers strong interpretability and aligns well with the keyword-based structure of the data, SentenceTransformer demonstrated superior clustering performance, with a smaller proportion of job postings being assigned to the noise cluster (cluster −1) and the differentiation between job clusters being clearer.

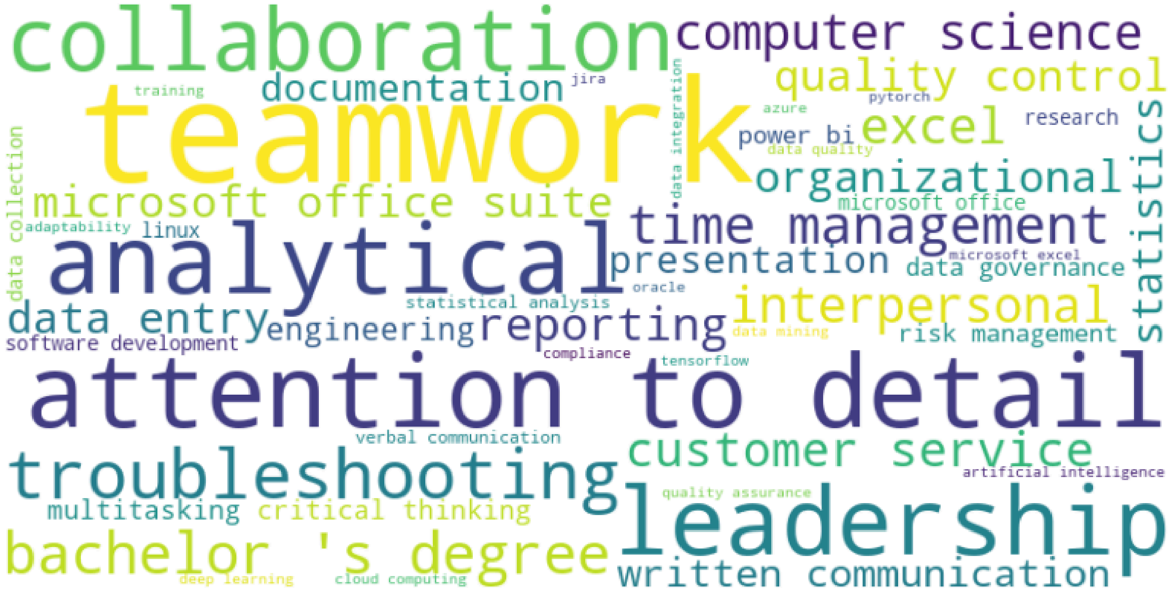
3.3 Clustering & Interpretation of cluster result

HDBSCAN was selected as the clustering algorithm due to its strong performance in prior job market segmentation studies [4]. The min_cluster_size hyperparameter was set to 50 after testing, as this value produced the most stable and interpretable cluster structure.

The final model identified three clusters, with approximately 50% of observations assigned to cluster −1 (noise). This outcome suggests substantial overlap in skill requirements across many data science roles. Word-cloud were used to identify the distinctive skill requirements associated with each identified job category.

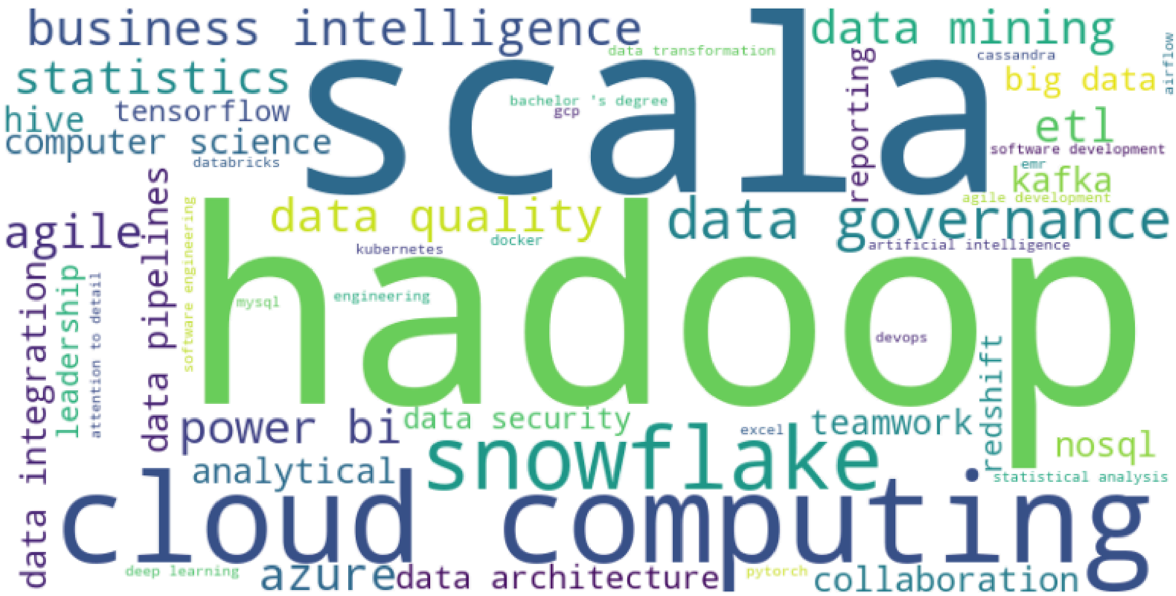
Cluster −1 is characterized primarily by soft skills such as *attention to detail*, *teamwork*, and *collaboration*. This cluster likely represents either entry-level roles with limited technical specialization or managerial roles where interpersonal and organizational skills are prioritized.

Cluster -1 – Skill Word Cloud



Cluster 1, accounting for approximately 49% of the data, corresponds strongly to data engineering and analytics infrastructure roles, indicating sustained market demand for these positions.

Cluster 1 – Skill Word Cloud



Although clusters 0 and 2 together represent only around 1% of the dataset, they are highly coherent, corresponding to medical data science roles and cybersecurity/data governance roles, respectively.

Cluster 0 – Skill Word Cloud

Cluster 2 – Skill Word Cloud

4. Findings, reflections and further work

4.1 Findings and reflections on the research questions

This project has successfully addressed the research questions. Specifically, Python, SQL, and communication skills are the most frequently mentioned requirements across job postings. Figure 1 presents the top 20 most common skills in data science, including both technical and soft skills. In addition, cluster −1, which accounts for approximately 50% of the dataset, highlights the importance of soft skills in data science roles. This reflects the interdisciplinary nature of the field, where collaboration and communication are essential alongside technical expertise.

- The combination of SentenceTransformer and HDBSCAN proved capable of dividing job postings into meaningful clusters. While SentenceTransformer is not perfectly suited to datasets dominated by technical terms, it still provides benefits when skill lists contain semantically related concepts. For clustering, HDBSCAN proved to be a robust method. With this dataset, setting min_cluster_size to 50 produced the most stable results. However, despite the model's ability to form clear clusters, approximately 50% of the data was assigned to cluster −1. This indicates that a large proportion of data science roles share similar and common skill sets, making them difficult to separate into distinct categories. From a practical perspective, this suggests that beginners in data science should prioritize acquiring core skills before specializing further to target specific roles.
- Finally, interpreting the four identified clusters provides useful insights into the skill requirements of different job categories. For infrastructure- and data engineering–focused roles, job seekers should emphasize skills such as Scala, Hadoop, and cloud computing. Data science roles related to medicine are also in demand; however, these positions typically require a background in medical science and relevant certifications such as ASCP in addition to core data science skills. For roles related to data governance, candidates should focus on generative AI, cybersecurity, and legal compliance related to data usage and management.

Regarding the dataset, although it successfully represents the US job market, the job_skills column was not fully standardized. Some irrelevant keywords remained and required manual filtering. Moreover, approximately 5% of the postings contained skill keywords but were not truly related to data science, which increased the complexity of the analysis. Nevertheless, given the dataset's size and the fact that over 90% of the data was usable, the analysis still provides reliable and meaningful insights overall.

4.2 Limitation & Future Work

Despite its strengths, the dataset has several limitations. As it reflects job postings from 2024, it may not capture the most recent market trends. Future work could involve collecting data directly from job platforms using APIs to ensure the most updated insights. Moreover, further analysis could categorize skills into soft and hard skill groups and segment roles by seniority level (e.g., junior, senior, or managerial) to create a cleaner and more structured dataset.

In terms of clustering, this study did not perform a systematic comparison of different method combinations using quantitative clustering evaluation metrics. Instead, model performance was primarily assessed based on the interpretability of clusters and the proportion of job postings assigned to the noise cluster (cluster −1). While this approach aligns with the exploratory nature of the analysis, it relies partly on qualitative judgment. Future work could leverage metrics such as the Davies–Bouldin Index or the Calinski–Harabasz score [4] to provide a more objective comparison between clustering methods and to strengthen the robustness of the conclusions.

5. References

[1] World Economic Forum, 2025. *The Future of Jobs Report 2025*. Geneva: World Economic Forum.

[2] Gunklach, J., Nadj, M., Michalczyk, S., Jacob, K. and Gröger, C., 2025. Beyond the unicorn? Job roles in data science. Business & Information Systems Engineering.

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[4] Lukauskas, M., Šarkauskaitė, V., Pilinkienė, V., Stundžienė, A., Grybauskas, A. and Bruneckienė, J., 2023. Enhancing skills demand understanding through job ad segmentation using NLP and clustering techniques. Applied sciences, 13(10), p.6119.

[5] Asaniczka. 2024. *Data Science Job Postings & Skills (2024). Kaggle*